

Ryan Faraji

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EDUCATION

Purdue University, Indianapolis, IN
Bachelor of Science in Mechanical Engineering

May 2028

RELEVANT EXPERIENCE

BoilerSub, Structures Team Member

November 2024 - Present

- Led the mechanical design and prototyping of the torpedo system in NX and SolidWorks for an Autonomous Underwater Vehicle (AUV) competing in an international robotics competition
- Used FEA and CFD to evaluate torpedo performance and collaborated with teammates to optimize mechanical components for underwater performance

Layered Materials & Structures Lab, Intern

May 2026 - July 2026

- Researched high-entropy MAX phase synthesis, developing a non-equimolar six-metal composition by powder metallurgy techniques
- Synthesized 2D transition metal carbides (MXenes) from MAX phase precursors through chemical etching
- Increased sample recovery by designing and fabricating a custom mortar cap to contain powder scatter during the grinding process

Purdue Space Program (PSP), Manufacturing Team Member

August 2025 - May 2026

- Designed and iterated a body tube roller using SolidWorks to produce rocket body tubes and support composite layups
- Managed CAD revisions and coordinated design updates with the PSP team to support fabrication and prototyping

Institute of Electrical and Electronics Engineers (IEEE), CAD Team Member

November 2024 - May 2025

- Designed a gantry structure for a resistor-sorting robot using Fusion 360 to improve mechanical performance
- Built and soldered circuit boards to connect electrical systems with mechanical components

Motorsports Engineering Team, Aerodynamics Team Member

September 2024 - May 2025

- Modeled aerodynamic assemblies in SolidWorks to improve go-kart performance through precise 3D geometry
- Simulated airflow and analyzed fluid dynamics in STAR-CCM+ to validate and refine component effectiveness

LEADERSHIP EXPERIENCE

Purdue Mechanical Engineering Ambassadors (PMEA), Secretary

October 2025 - Present

- Maintained sign-up spreadsheet and ensured each event/tour had appropriate PMEA representation
- Ensured seamless onboarding of new members and sufficient engagement from all Ambassadors

Middle Eastern Engineering Society (MEES), President & Founder

August 2025 - May 2026

- Led a student organization focused on building community, professional development, and networking opportunities for Middle Eastern engineering students
- Secured \$5,000 to support hands-on competitions and organized speaker events, growing club presence on campus

EPICS, Project Manager

August 2024 - May 2025

- Led a group of engineering students to create an RC car workshop for prospective students
- Integrated circuitry and Arduino programming to enable functional control of an RC vehicle
- Oversaw scheduling of tasks and weekly accomplishments by the class and sub-teams

TECHNICAL SKILLS & CERTIFICATIONS

Certifications: OSHA 10-hour

Manufacturing: Soldering, Welding, Woodworking

CAD & CAM: NX, SolidWorks, Fusion 360

FEA & CFD: SolidWorks Simulation, SolidWorks Flow Simulation, Star-CCM+

Project & Communication: Technical Writing, Microsoft Office Suite, Project Scheduling